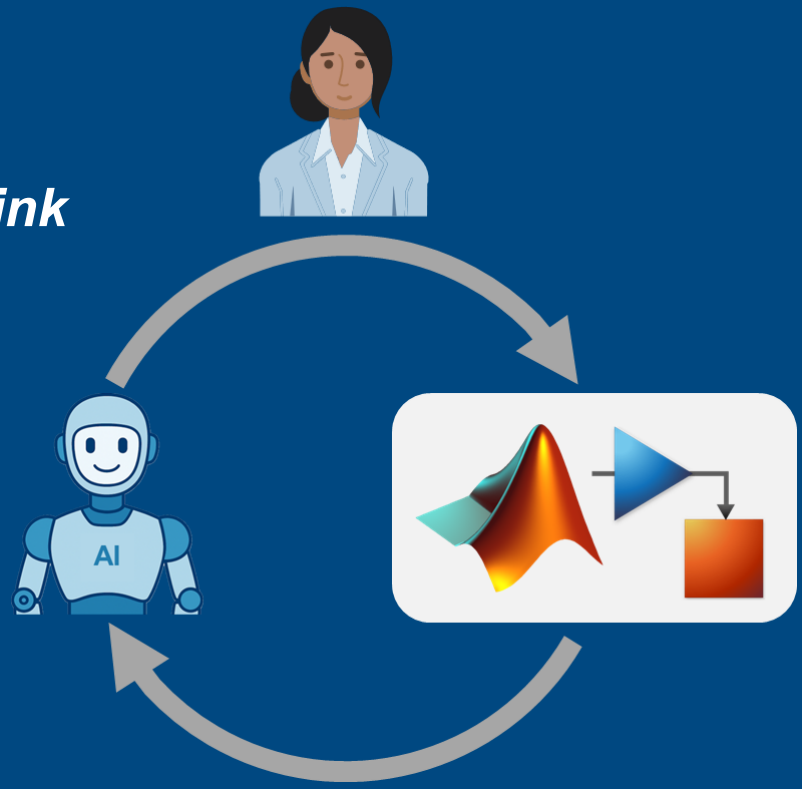
MILCOM 2027 — Tutorial TUT13
AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

Part 1 — Agentic AI Workflows with MATLAB and Simulink



Sekhar Sekharan– sekhars@mathworks.com

Applications Engineer
MathWorks | RF/EM Group

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Agentic AI helps engineers choose antenna designs

The goal is a defensible trade-off, supported by EM simulation.

THE CHALLENGE

Footprint, bandwidth, gain
Repeated setup and checking
Many possible design choices

WHAT THE AGENT CHANGES

Reads requirements and plans
Runs tools and inspects results
Chooses the next experiment

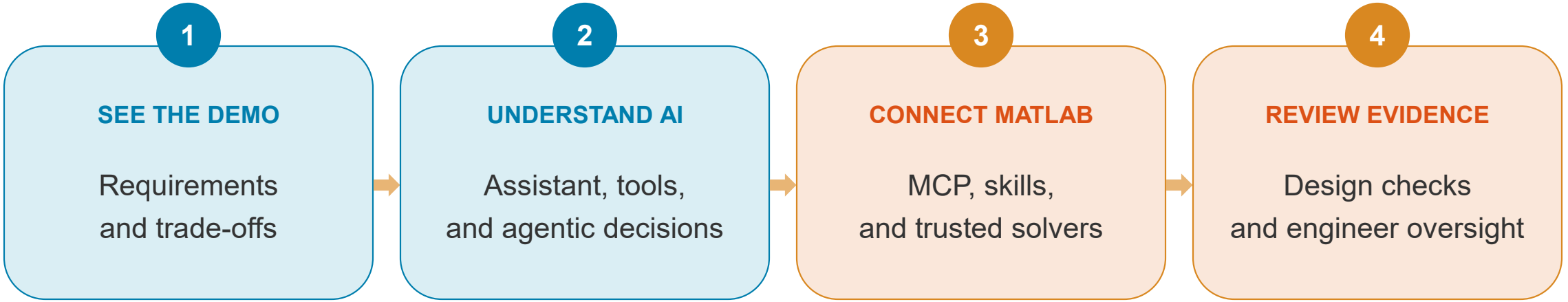
WHAT YOU WILL SEE

Candidate comparison
RF checks in MATLAB
Evidence for engineer review

Requirements > Experiments > EM evidence > Engineering decision

Our journey: from an antenna request to engineering evidence

One antenna problem connects the demonstration to the concepts.



PART 1 SCOPE

Antenna EM modeling	Tools and domain skills	Engineering decisions

The specification demo is the baseline; the new trade-off study extends it into an engineering loop.

Predictive models and agents play complementary roles

One predicts a result; the other coordinates the work needed to make a decision.

PREDICTIVE AI MODELS

Predict antenna performance
Reconstruct radiation patterns
Use trained, task-specific models

PART 2 • Prediction and reconstruction

AGENTIC WORKFLOWS

Plan and coordinate experiments
Call numerical tools and models
Revise the plan using results

PART 1 • The engineering workflow

An agent can call a predictive model.
EM simulation checks the selected designs.

Agentic work adapts the next step to the evidence

Tool access enables action. Planning, feedback, and revision close the loop.

AI ASSISTANT	TOOL-ASSISTED TASK	AGENTIC WORKFLOW
Explains antenna concepts Drafts MATLAB code Engineer chooses each step	Runs a requested analysis Returns plots and metrics Follows the specified sequence	Plans, executes, evaluates Revises candidate experiments Reports evidence and limits
“Explain this S11 curve.”	“Compute S-parameters for this antenna.”	**“Find a smaller design that still meets the RF targets.”**

PART 1 FOCUS: THE AGENT COORDINATES THE LOOP

The engineer sets constraints, reviews evidence, and accepts the design.

Proposed demo: choose a compact 2.45 GHz patch

Start from requirements, compare candidates, and justify the design choice.

DRAFT REQUIREMENTS

Band	2.40–2.50 GHz
Input match	S11 ≤ −10 dB
Broadside gain	≥ 5 dBi, realized
Board outline	≤ 70 × 70 mm

RF targets apply across the band.

TRADE-OFF AND DESIGN CHOICES

Smaller board ↔ More RF margin

Patch dimensions and feed geometry

Finite ground-plane dimensions

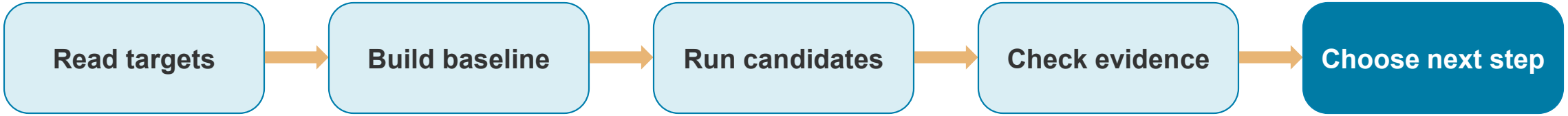
Two permitted substrate thicknesses

Deliverable: feasible candidates, an area–gain trade-off, and a reproducible design recommendation.

Proposed targets for EM rehearsal. Feasibility and performance have not yet been established.

The demo will expose decisions, tools, and evidence

The agent chooses the next experiment; MATLAB evaluates the candidate.



Repeat within the budget: choose a permitted change, rerun, and record why.

PROMPT + DECISION LOG

What changed?
Why run this experiment?

SHORT MATLAB EVALUATOR

Geometry → solve → metrics
Versioned, rerunnable code

PLOTS + REQUIREMENT TABLE

S11 + realized gain + area
Feasibility before ranking

Optional bridge to Part 3: a 2×2 array with mutual coupling.
Start at broadside; add beam steering after the coupled model is understood.

Baseline demo: antenna design from a specification

MATLAB R2026b

HOME PLOTS APPS

Search (Ctrl+Shift+Space)

Sekhar

New Script | New Live Script | New | Open | Go to File | Find Files — FILE

Import Data | Save Workspace | Clean Data | New Variable | Open Variable | Clear Workspace — VARIABLE

Favorites | Run and Time | Clear Commands — CODE

Simulink — SIMULINK

Layout | Set Path | Add-Ons — ENVIRONMENT

Help | Request Support | Feedback | Learn MATLAB — RESOURCES

Copilot Chat — COPILOT

C: › Users › sekhars › OneDrive - MathWorks › Workshop_AI_Antenna_MATLAB_Copy › Demos › Part1_AgenticAI › Demo_Start

Files

Name

antenna_demo_requirements.md

Workspace

Name	Value	Size
ans	*1×1 terminal*	1×1
mcpServer	"C:\MCP\VE...	1×1
satkRoot	"C:\Users\se...	1×1

Editor

ArrayinBox_pcbStack.m | ArrayinBox_pcbStack_refinement.m

C:\Users\sekhars\OneDrive - MathWorks\Alt_Customer\internal_mesher_issue_aram\Ar...

```
% Square-feed PCB antenna: original S11 mesh ref

%% Antenna
f0 = 1.67e9;
lambda = 3e8/f0;
d = dielectric("FR4");
pm = patchMicrostrip(Length=75e-3, Width=37e-3,
    GroundPlaneLength=120e-3, GroundPlaneWidth=8
pm = design(pm, f0);
% pm = pcbStack(pm);
% pm.FeedViaModel = 'square';

%% Box
boxLength = 0.2;
```

Apps

Terminal

```
C:\Users\sekhars\OneDrive - MathWorks\Workshop_AI_Antenna_MATLA
B_Copy\Demos\Part1_AgenticAI\Demo_Start>devai launch codex
devai 1.26.0 is available (you have 1.24.1). Update now? [Y/n]

Downloading devai 1.26.0...
Checking for latest release...
Latest version: 1.26.0-windows-amd64
Downloading update...
Updated to 1.26.0
devai updated to 1.26.0.
Please rerun: devai launch codex

C:\Users\sekhars\OneDrive - MathWorks\Workshop_AI_Antenna_MATLA
B_Copy\Demos\Part1_AgenticAI\Demo_Start>
```

Command Window

```
>> terminal

ans =

  terminal with properties:

    Shell: "C:\Windows\system32\cmd.exe"
    Place: "matlab"
    Theme: "auto"

>>
```

mathworks-my.sharepoint.com is sharing your screen. Stop sharing Hide

Search

5:32 PM
10/11/2026

MATLAB R2026b

HOME PLOTS APPS EDITOR PUBLISH VIEW

Search (Ctrl+Shift+Space)

Sekhar

New Open Save Print Compare — FILE

Go To Find Bookmark — NAVIGATE

Copilot Refactor — CODE

Profiler Code Issues Debugger — ANALYZE

Generate Test Find Tests — TEST

Run Section Section Break Run and Advance Run to End — SECTION

Run Step Stop — RUN

C: › Users › sekhars › OneDrive - MathWorks › Workshop_AI_Antenna_MATLAB_Copy › Demos › Part1_AgenticAI › Demo_Start

Files

Name

- baseline_results
- antenna_demo_requirements.md
- plotBaselineAntenna.m
- runBaselineAntenna.m
- writeBaselineReport.m

Workspace

Name	Value	Size
ans	1×1 terminal	1×1
baseline	1×1 struct	1×1
baselineFigures	1×3 Figure	1×3
baselineSaved	1×1 struct	1×1
mcpServer	"C:\MCP\VE...	1×1
satkRoot	"C:\Users\se...	1×1

Editor

_pcbStack.m | ArrayinBox_pcbStack_refinement.m | untitled

1 Press Ctrl + Shift + P to generate code with Co

Apps

Terminal

```
› create a baseline model and evaluate its s11 using a fast i…

  Interpolation passed direct checks at both band edges and
  center: maximum complex-S error 0.000022, below 0.01.

  View saved figures: Antenna (baseline_results/
  baseline_geometry.png) · S11 (baseline_results/
  baseline_s11.png) · 3D directivity (baseline_results/
  baseline_directivity_3d.png).

  Source: runBaselineAntenna.m. Model, numerical data, and
  figures are saved under baseline_results. Run
  plotBaselineAntenna to redisplay without solving again.

  These use the λ/16 screening mesh. Directivity does not
  establish the realized-gain requirement.

  Worked for 3m 19s • 5:39 PM


› Ask Codex to do anything

  GPT-6-Astra high · ~\OneDrive - MathWorks\Workshop_AI_Antenn…
  ? for shortcuts                    ⚠1 warning · f2 to view
```

Command Window

```
    Theme: "auto"

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Editor: 100% UTF-8 CRLF Script Ln 1 Col 1

Search

5:40 PM
10/11/2026

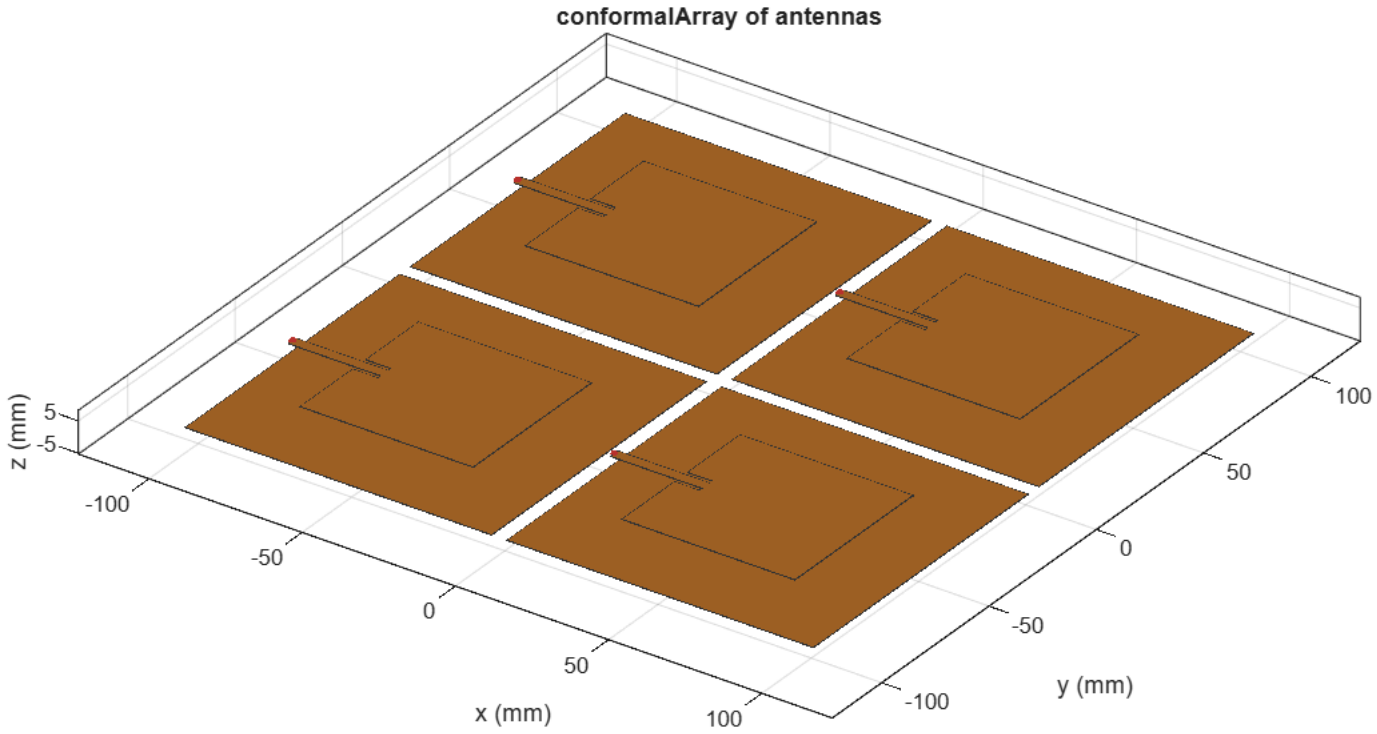
The selected patch becomes a coupled 2x2 array

After the baseline and bounded test, continue from the selected D6 design.

PROMPT ENTERED

Extend the chosen design to a 2x2 array. Include coupling and recommend element spacing.

Same D6 radiator
Four 100 x 100 mm boards
Equal, in-phase excitation



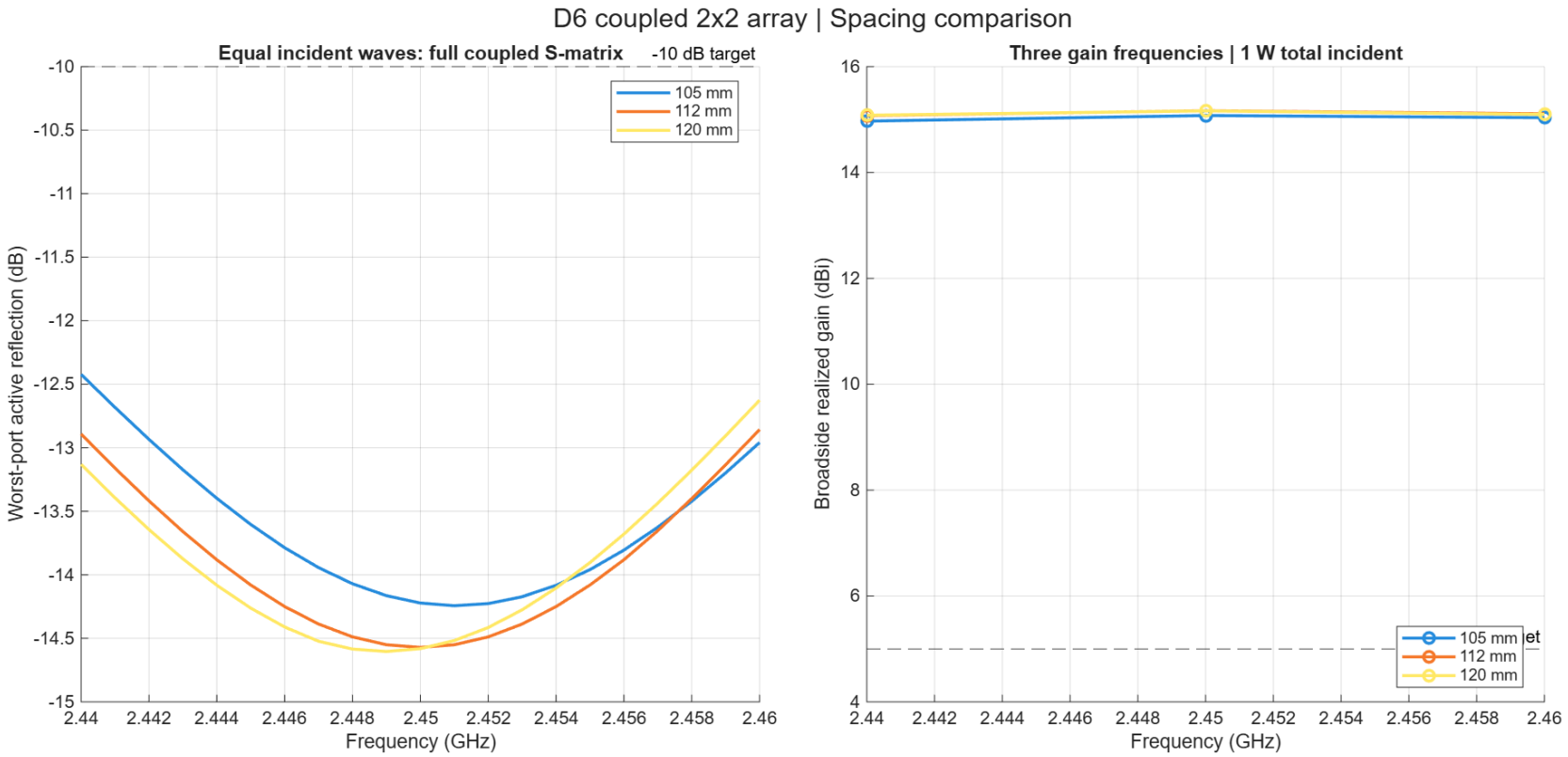
Obtained: four coupled elements, 105 mm center spacing, 5 mm board gaps.

The agent carries the selected design into the next engineering question.

Source: Demo_Start/array_results | saved study, 12 Oct 2026 | antenna-only screening

105 mm spacing keeps almost all the sampled gain

Result obtained: compare 105, 112 and 120 mm using the same coupled EM model and mesh.



105 mm

Provisional recommendation
205 x 205 mm array outline

14.97 dBi

Minimum broadside gain
at the three sampled
frequencies

Only 0.11 dB

more gain at 120 mm;
outline grows to 220 x 220 mm

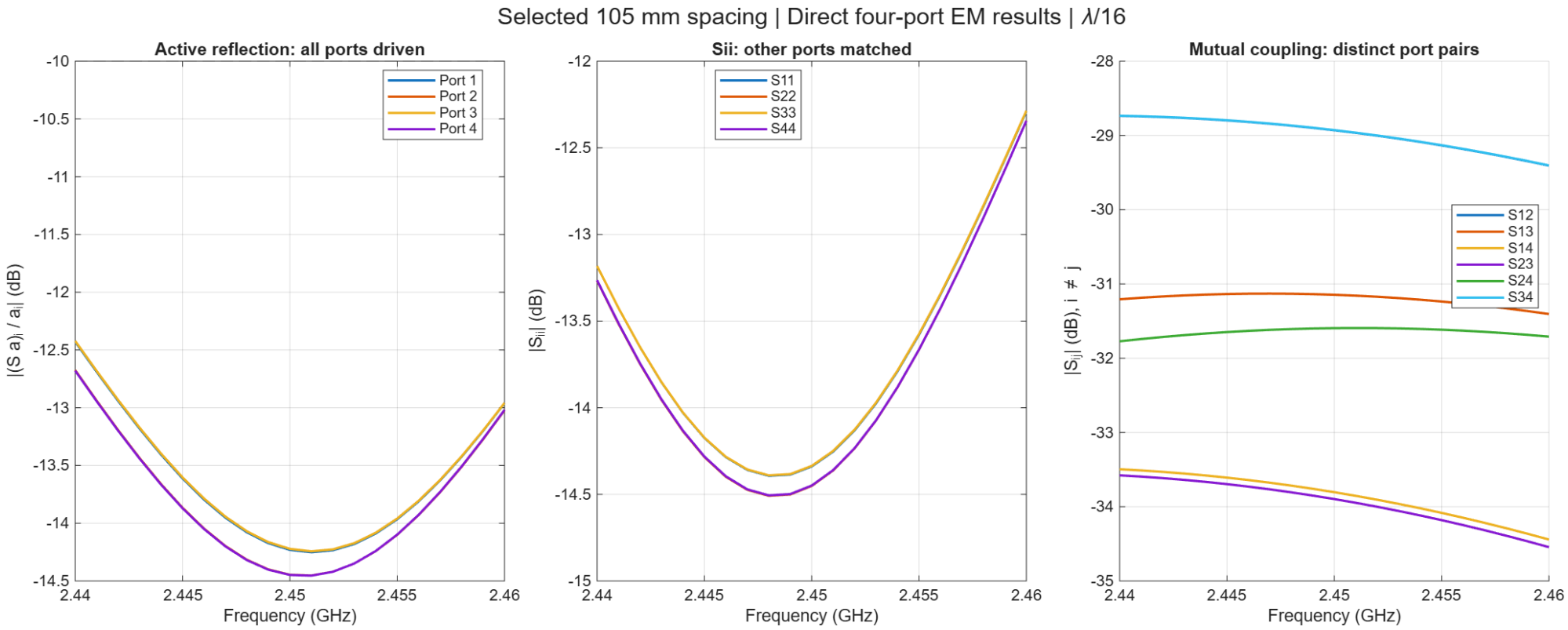
All three pass the sampled screening targets: active reflection <= -10 dB and broadside gain >= 5 dBi.

The agent turns a sweep into a recommendation supported by a size-versus-gain trade-off.

Source: spacing_comparison.csv and array_spacing_comparison.png | lambda/16 mesh | 2.440-2.460 GHz

Coupling is included; final verification is still open

Worst active reflection: -12.42 dB. Strongest coupling: -28.73 dB. Physical feed losses excluded.



Pending: finer mesh (17/21 S samples; 0/3 gain samples), 3D pattern, and remaining field checks.

The agent preserves evidence and identifies the checks needed before accepting a design.

Source: array_port_curves.png and array_summary.md | direct four-port EM solves | selected 105 mm spacing

The array result points to the next feed experiment

Next question: can a physical feed retain at least 14 dBi from one 50-ohm input?

FOLLOW-UP PROMPT

Give me a table of candidates to test, including modeling the feed lines in MATLAB. Help me plan candidates to achieve an array goal.

OBTAINED: SIX CANDIDATES + MATLAB LINE TEMPLATES

Test pair	Controlled change	Purpose
F1 -> F4	Low-loss Wilkinson: 100 -> 60 mm routes	Test routing loss first
F2 / F3	FR4 / matched-tee alternatives	Compare material / topology
F5 / F6	112 / 120 mm center spacing	Spend area only if useful

PROPOSED LOADED GOALS

Gain >= 14 dBi
Input S11 <= -10 dB
Amplitude spread <= 0.5 dB
Phase spread <= 5 degrees

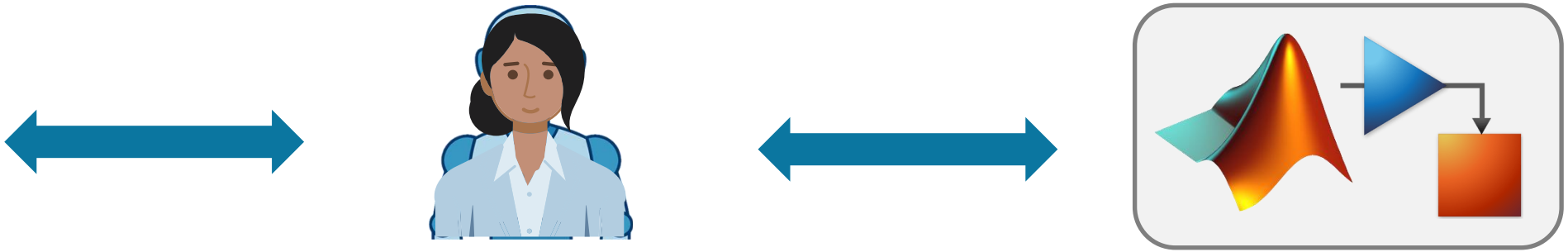
Plan and initial line dimensions saved.
Feed performance not yet evaluated.

Next: finish array verification, then test the low-loss feed and shorter equal-length routes.

The agent connects results to the next test plan; the engineer sets targets and accepts the evidence.

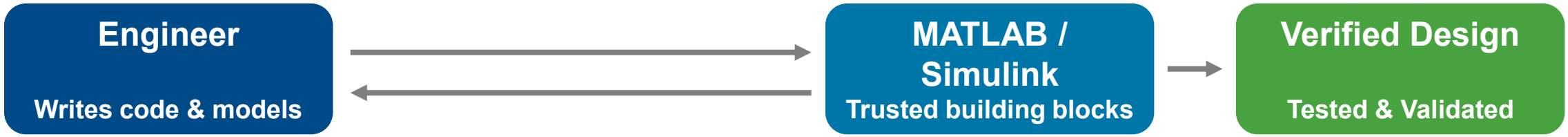
Source: Demo_Start/feed_plan/array_feed_plan.md and feed_candidates.csv | proposed goals; all six candidates planned

Recap : How is Generative AI is transforming engineering

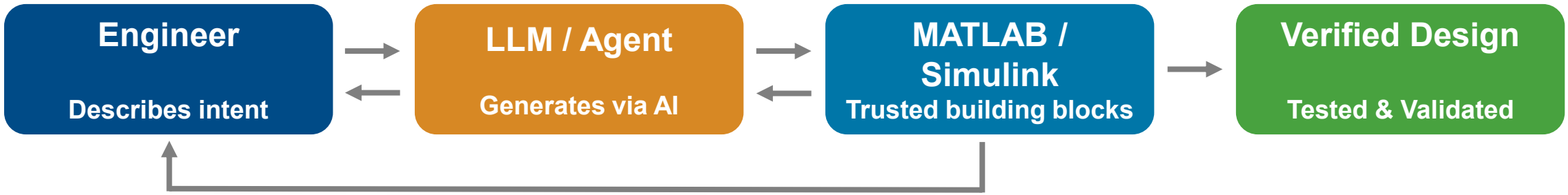


GenAI changes how you create - not engineering accountability

Today



With AI



What doesn't change when you add AI

The engineer owns the design. The same engineering evidence verifies it.

Accelerate your MATLAB and Simulink productivity with GenAI

Agentic Workflows



External GenAI Agents

use MATLAB or Simulink with external Agents via MCP or VS Code



Build Agents in MATLAB

call LLMs from MATLAB code

AI Assistants



MATLAB and Simulink Copilots

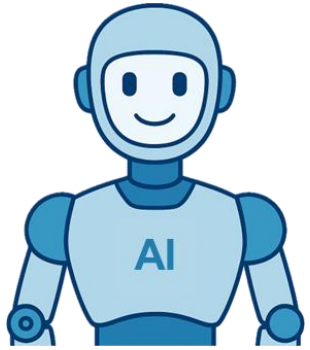
optimized for MATLAB and Simulink



External GenAI Agents

use MATLAB or Simulink with external Agents via MCP or VS Code

Agentic workflows are enabled by Model Context Protocol (MCP)



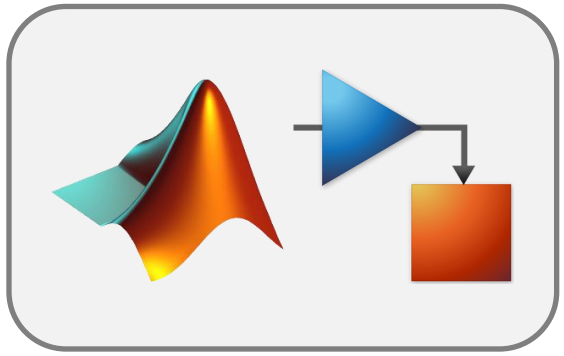
Cloud providers
OR
Local models

MCP Client

MCP



MCP Server



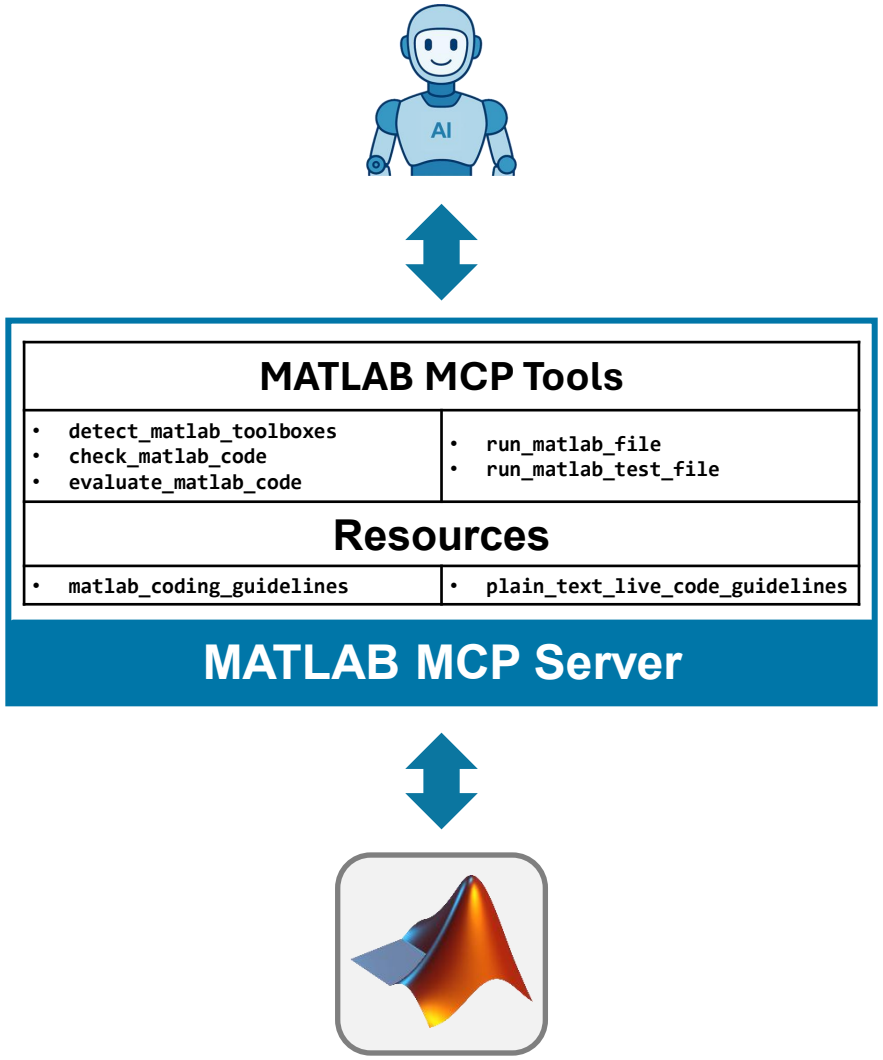
Tools

MATLAB MCP Server

Enabling GenAI Agentic Workflows with MATLAB

A **free, open-source** local MCP server that connects GenAI assistants to a **locally** licensed MATLAB, enabling intelligent automation and code execution

- No additional license required – just connect and go
- Preserves privacy with local execution
- Start a new MATLAB session or connect to an existing session



https://github.com/matlab/matlab-mcp-server

How can we improve agent behavior?

Skills transform prompts into reliable, domain-aware, reusable workflows

- Right knowledge at the right time
- Reliability and output quality
  - Encoding proven procedures
  - Grounding against hallucination
- Knowledge the model may not have
  - Domain and proprietary expertise
  - Institutional memory

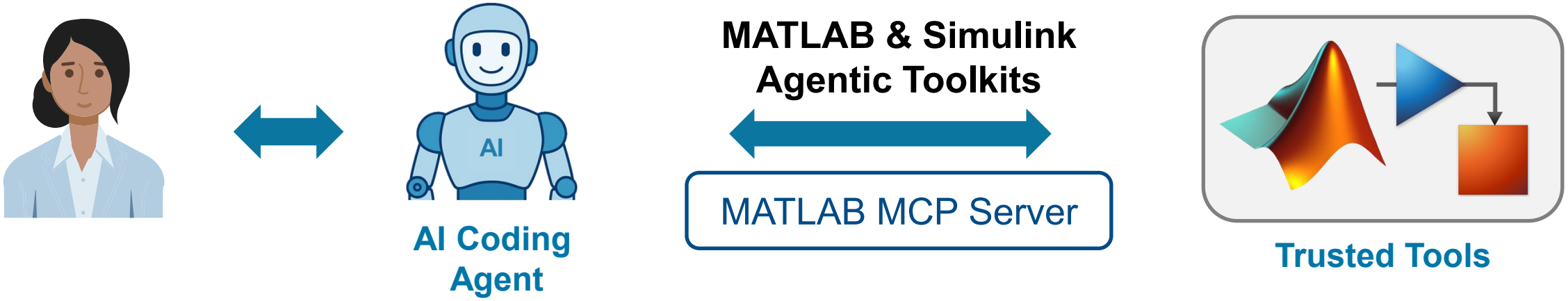
/skills/skill-name/SKILL.md

```
---
name: skill-name
description: Defines the conditions under which
the skill should or should not be triggered
---


# Instructions
This section is loaded by the agent only if it
determines, based on the skill description, that
the content may be relevant to the current task.
```

Skills tells your agent how to do things correctly the first time around

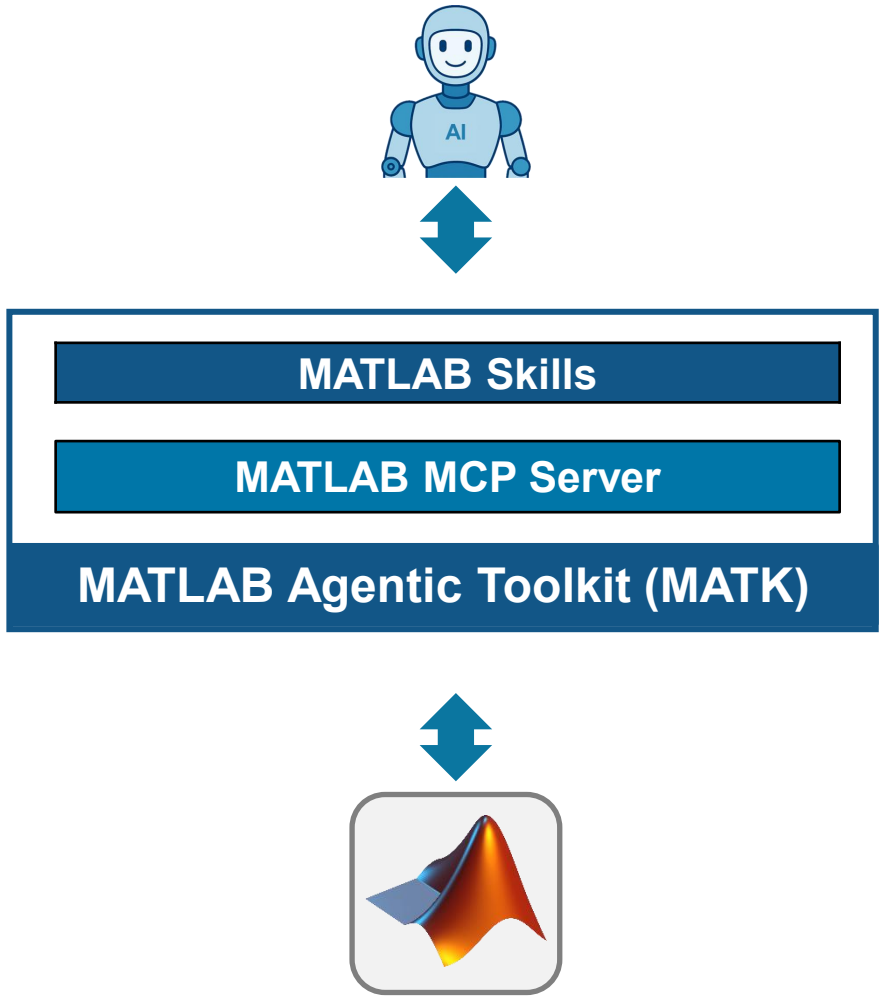
MATLAB & Simulink Agentic Toolkits: Bring Trusted Capabilities to AI Agents



MATLAB Agentic Toolkit

Connect trusted MATLAB capabilities to AI agents to make engineering and scientific workflows agent-ready

- MATLAB MCP Server
- Expert Skills Catalog encode MATLAB best practices, guiding AI in testing, debugging, and code modernization.
  - MATLAB Software Development
  - Signal Processing
  - Image Processing and Computer Vision
  - Wireless Communications
  - .. and many more!!



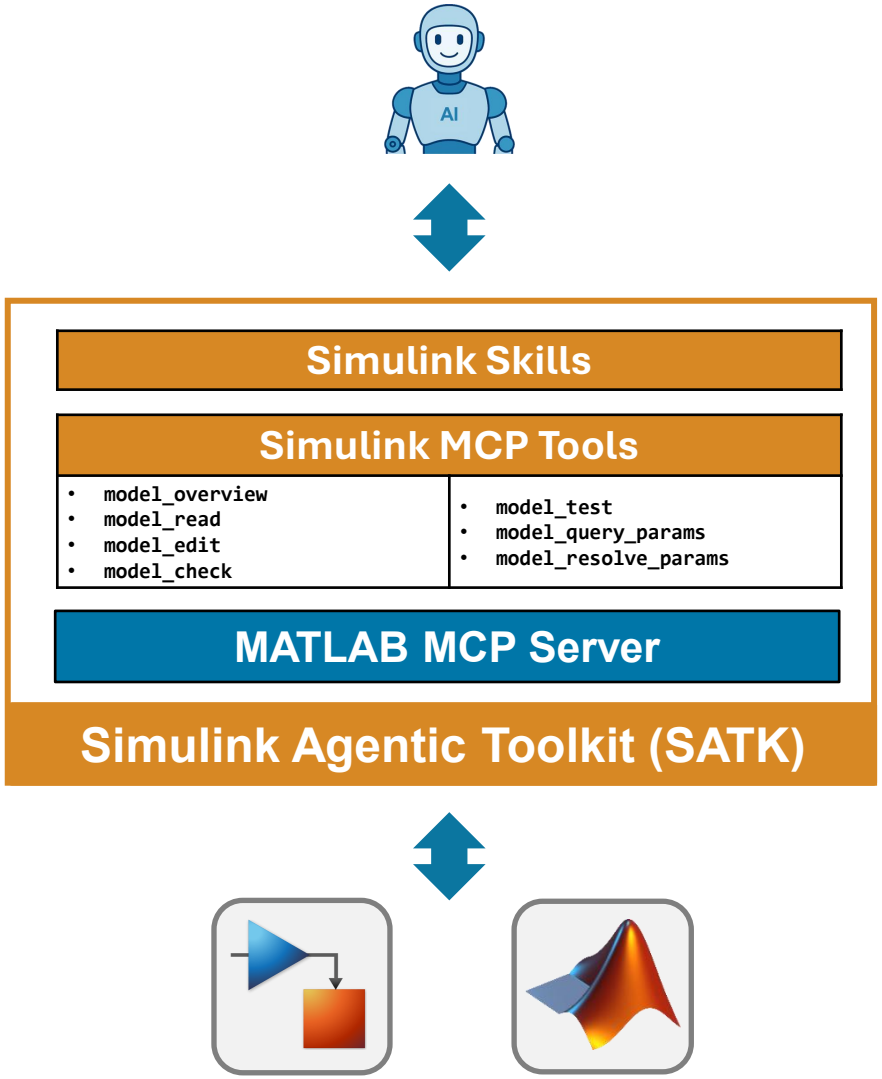
https://github.com/matlab/matlab-agentic-toolkit

Simulink Agentic Toolkit

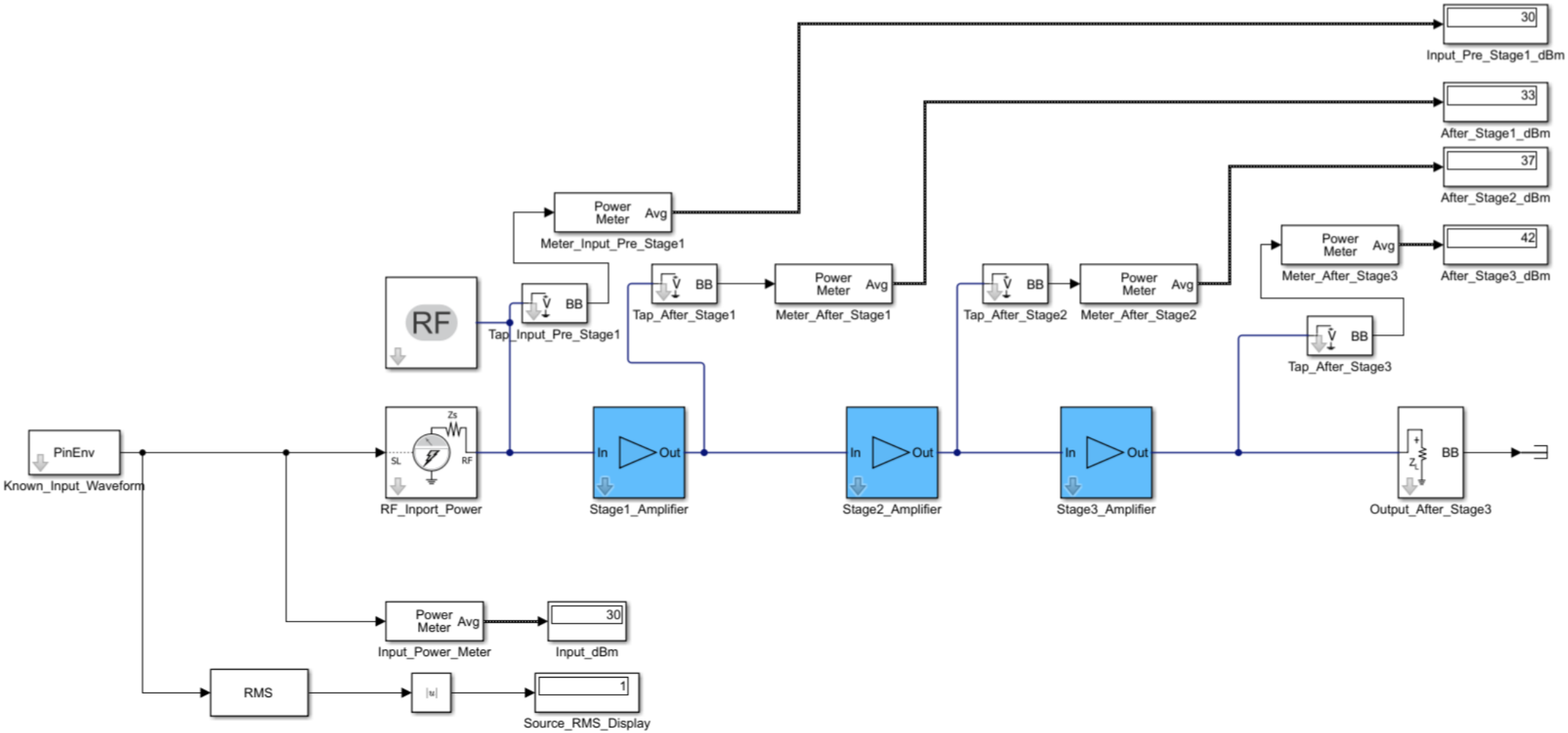
Connect agents to Simulink giving them the ability and the knowledge to work with Simulink models effectively.

- Builds on top of the MATLAB MCP Server
- MCP tools optimized for reading, editing, querying, testing, and checking Simulink models
- Agent skills encoding Model Based Design best practices for model building, plant specification, testing, requirements, and architecture modeling

https://github.com/matlab/simulink-agentic-toolkit

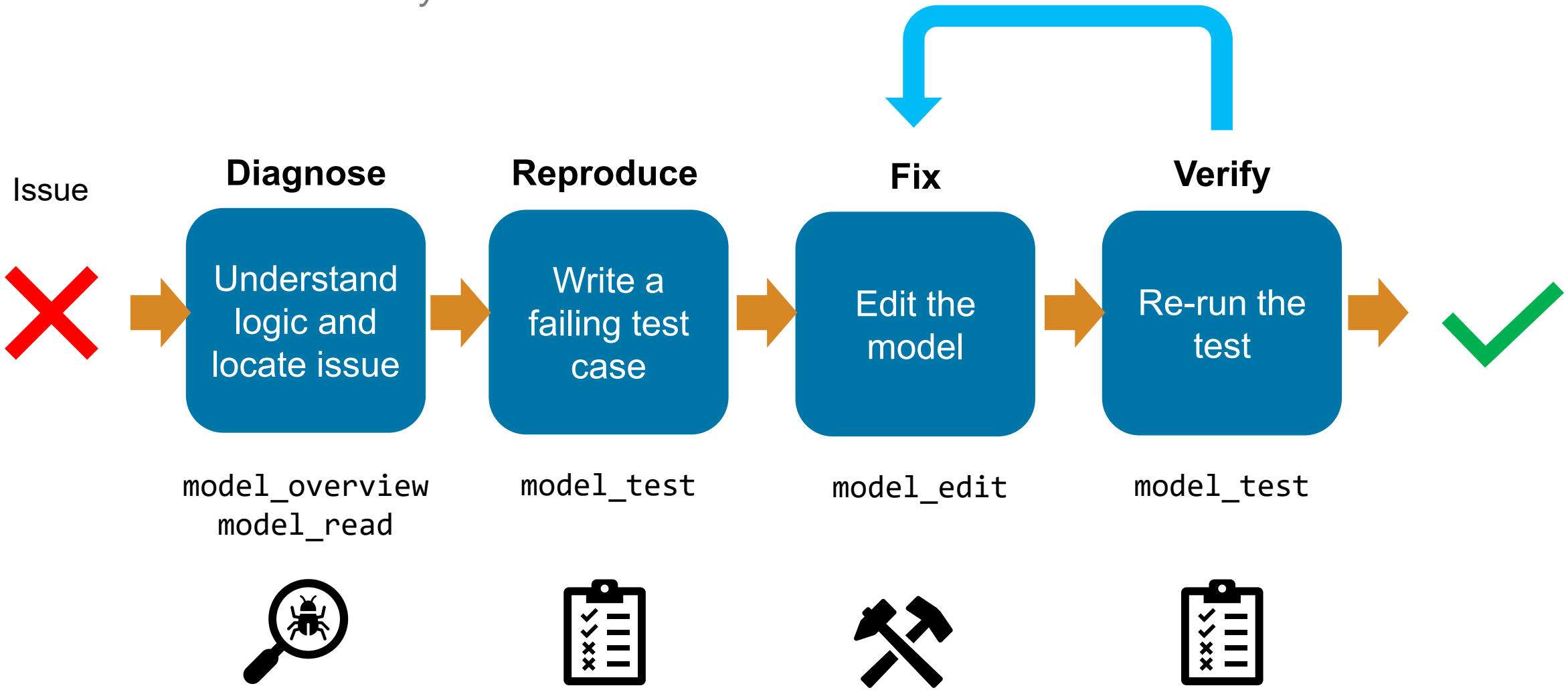


Example RF Model Generated with Agentic Workflow



Simulink Agentic Toolkit enables automation of complex workflows

Troubleshoot – Fix – Verify



Other offerings

- Build custom agents in MATLAB
- Deploy your MATLAB algorithms and models as MCP tools
- Polyspace Agentic Toolkit
- MATLAB Copilot
- Simulink Copilot
- Polyspace Copilot

Key Takeaways

- Agentic AI is powerful; with the right tools it is transformational.
- Regardless of your AI tools, there is a way to use them with MathWorks Tools
- Stay informed about new updates: https://www.mathworks.com/products/matlab/agentic-ai.html
- Ready to get started? Schedule a follow up to discuss workflow challenges and explore new ideas

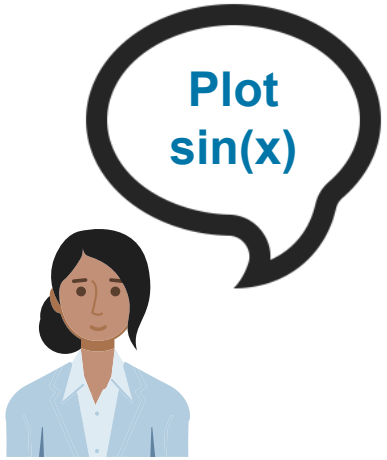
Resources

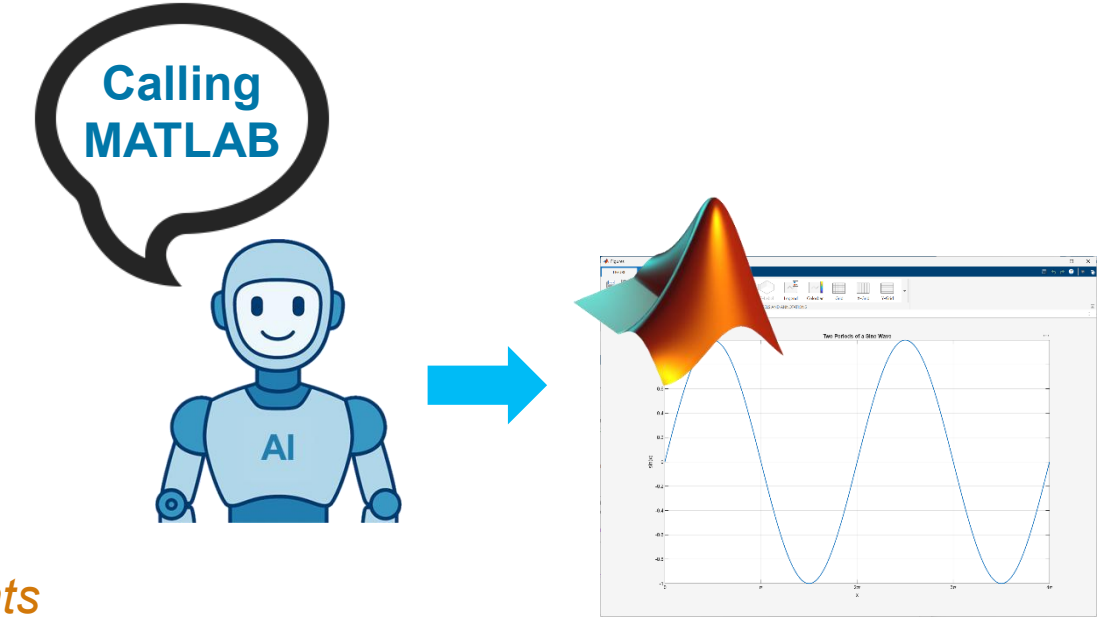
- Generative AI with MATLAB and Simulink (Overview)
- MATLAB MCP Server
- MATLAB Agentic Toolkit (MATLAB MCP Server + MATLAB Skills)
- Simulink Agentic Toolkit (MATLAB MCP Server + Simulink Tools/Skills)

Accelerating Design and Analysis in MATLAB Using Agentic AI

After this 1-day training you will be able to:

- Explain how agents operate and differ from past tools
- Connect MATLAB with AI agents
- Work with agentic AI to create MATLAB code and accelerate complex tasks
- Understand, debug, and extend large code bases in MATLAB with agentic AI assistance
- Leverage domain expertise by creating custom skills for repeatable workflows





Note: this training is available only for private training events

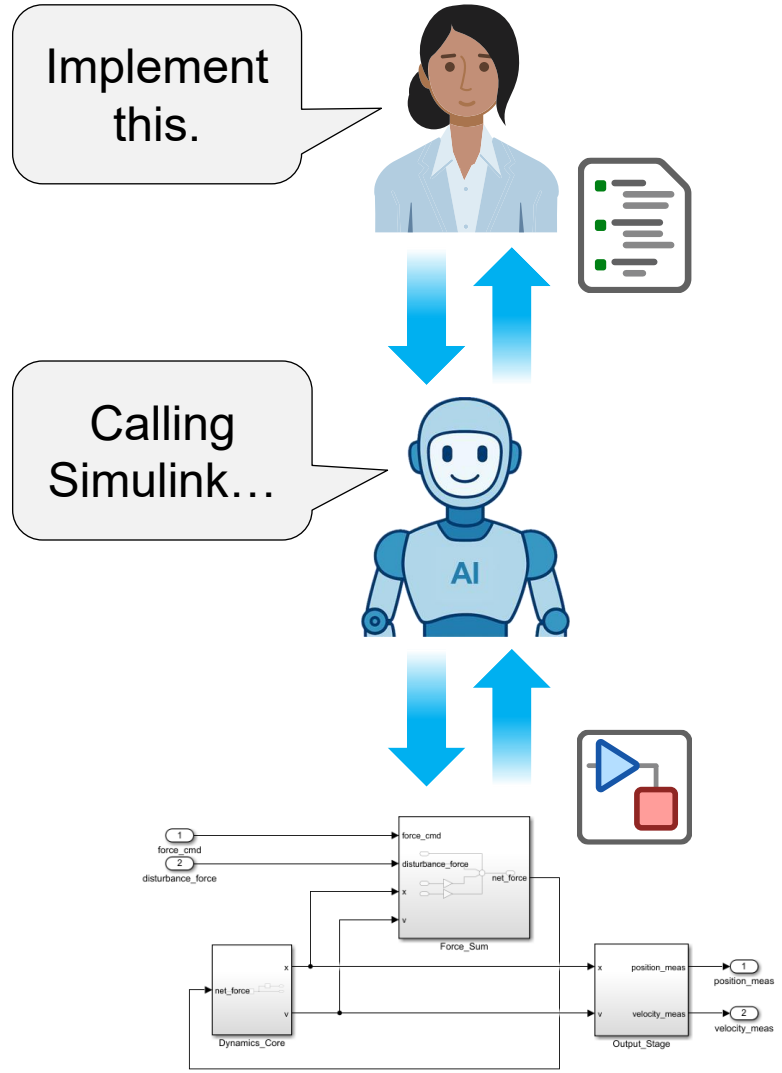
Accelerating Engineering Design in Simulink Using Agentic AI

After this 1-day training, you will be able to:

- Recognize how agentic AI systems use context, tools, and review loops to support Simulink engineering tasks.
- Build, inspect, and edit basic models using the Simulink Agentic Toolkit.
- Use agentic AI to complete a Simulink engineering task with review checkpoints, simulation, testing, and documentation.
- Improve consistency and incorporate domain knowledge for Simulink workflows using custom skills.

Full Course Outline

Note: this training is available only for private training events



MathWorks Engineering Support



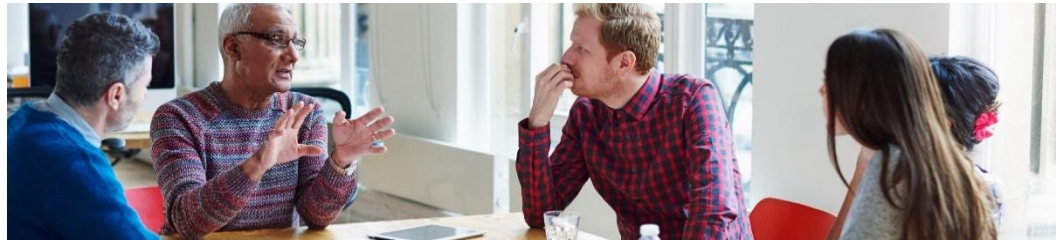
Training



Guided Evaluations



Onsite Workshops



Consulting



Technical Support

Questions?